KANCHANA GAYATHRI ARIYASINGHE

B.Sc.Eng (Hons) in Computer Engineering - Undergraduate

Location: No 31/6, Ranjan Wijerathna Mw, Rattota.
Mobile: +94 76 025 1767
Email: ariyasinghekanchana@gmail.com
LinkedIn: kanchanaAriyasinghe
GitHub: KanchanaAriyasinghe
My Profile: my_profile

EDUCATION

University of Jaffna — **Sri Lanka**
B.Sc.Eng (Hons) in Computer Engineering (Undergraduate) — April 2022 - Present
Cumulative GPA: 3.52 / 4.00 (After 5 Semesters)

SKILLS SUMMARY

- **Programming Languages**: Java, C++, Python
- **Frameworks:** React, Node.js, Spring Boot, Next.js
- **Databases:** MYSQL, MongoDB, PostgreSQL
- **Embedded Systems**: Microcontroller Programming
- **Tools & Frameworks**: Git, Jira, BitBucket
- **Problem Solving**
- **Teamwork**
- **Languages**: Sinhala (Native), English

WORK EXPERIENCE

Software Engineer Intern – Sri Lanka Telecom — **November 2025 – Present (6 months)**

- Working as a full stack developer on the Business Continuity Management System (BCMS) using the MERN stack (MongoDB, Express.js, React, Node.js).
- Working as a full stack developer to design and implement a Learning Path Management System (LPMS) for the HR Talent Development section using the PERN stack (PostgreSQL, Express.js, React, Node.js).

PROJECTS

Potholes Detection using Advanced Object Detection Techniques (Ongoing) — **January 2025 (Present)**
Researcher

- Developed a real-time pothole detection system using YOLOv8m and MobileNetV2, achieving 90.3% accuracy in low-light, rainy, and cluttered conditions.

Hotel Management System (Ongoing Project) — **November 2025**

- Developed a full-stack hotel management system with customer features for browsing hotel details, viewing gallery images, and making online room bookings (MERN stack).
- Implemented role-based functionalities for Receptionist (booking management), Housekeeping (room cleaning, maintenance, and status updates), and Admin (managing rooms, staff, and overall operations).
- Built and integrated RESTful APIs with MongoDB to ensure efficient data handling and seamless system performance.

I - Computers Website (Ongoing Project) — **January 2026**

- Developing a complete web application for I Computers using MERN stack, covering both frontend and backend functionalities.
- Creating responsive interfaces, managing data flow, and implementing RESTful APIs for smooth user and business operations.
- Handling authentication, database design, and performance improvements to ensure a secure and efficient system.

Computer Lab Booking and Equipments Borrowing System — **January 2025**
Role: Project Manager and Developer, Software Engineering
• Developed a full-stack web application named LabMate for computer lab booking and equipment borrowing using the MERN stack, Vite, Tailwind CSS, and JWT for secure authentication. Led the project through Jira-based sprint management, designed the admin frontend with React.js, and contributed to backend development with Node.js and MongoDB for efficient, real-time data handling.

Research AI — **November 2025**
• Developed intelligent research assistant chatbot that analyzes PDFs, discovers relevant academic papers and datasets, and provides AI-powered answers using Groq, integrating arXiv, CrossRef, and Hugging Face APIs with a Flask backend and a JavaScript-based frontend.

The Smart Medicine Reminder Box
• Developed the Smart Medicine Reminder Box, an IoT-based device featuring automated medication tracking, real-time scheduling via a Firebase-powered web interface, an LCD display with servo-controlled dual compartments for organized dispensing, and manual refill functionality for user convenience and reliability.

Student Feedback Management System — **January 2024**
• Developed a Student Feedback Management System to efficiently collect and manage course and lecturer feedback within the faculty, featuring secure authentication for students, lecturers, and administrators, and built using **PHP**, HTML/CSS, and MySQL with phpMyAdmin for robust data management.

Image Caption Generator using CNN-LSTM — **April 2025**
• Built a deep learning model to generate captions for images using pre-trained VGG16 for feature extraction and LSTM for text generation. Tools: Python, TensorFlow, Keras, NLTK, Google Colab

CPU Scheduling Algorithms Implementation — **January 2025**
• Developed a web-based system using React.js, JavaScript, and CSS to simulate various CPU scheduling algorithms, including FCFS, Round Robin, SPN, SRTN, and Priority Scheduling, with performance comparison based on average waiting and turnaround times.

Food Recipe Web Application — **November 2025**
Individual Project
• Developed an interactive Food Recipe Web Application using the MERN stack, enabling users to explore, create, and manage recipes, enhancing the cooking experience through a simple and engaging interface.

CERTIFICATION & COURSES
• Java Programming Internship (One month) – Codesoft
• AlgoXplore 1.0 Algorithm and Capture the Flag (CTF) Challenge - NSBM
• MoraXtreme 9.0, a 12-hour coding marathon - IEEE UOM
• TECHNOVATION 2025 | SLT-MOBITEL – AI Innovation Challenge (Finalist) – SLT-MOBITEL

VOLUNTEERING
• IESL Student Chapter - University of Jaffna — Membership Number: S-29763
• IEEE Student Member — Member number: 10041145